Beyond Prediction: Prescriptive AI for Strategic Energy Capital Allocation

A Governed Decision Intelligence System for Risk, Returns, and Investment Committee Choices Across Oil, Gas, Power, Renewables, and Storage

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Is prediction enough to spend a $3.82 billion budget well?

No. Machine learning ranked the candidates better than a hand-built lookup, but did not produce a demonstrably better portfolio. The value lay past prediction: in pricing the tail, in optimizing under the company's own rules, and in a challenger that argues but never decides.

The decision · 1 January 2025

$3.82 billion to commit across forty candidates, using only what was public that day. No technology may take more than 40% of the budget.

20 Gulf of Mexico leases

Redevelop now: capital spent today, output up 20% after two years. They rarely lose money, but all sell the same oil, so they lose together.

20 ERCOT queue requests

Cheap to hold, expensive once the interconnection agreement is signed. Most never get there: of 2000–2019 capacity, 12.6% was operating by 2024 and 76.9% had withdrawn.

Four words used here

Book: the candidates funded together. **Worst-5% loss:** the loss to budget for in the one year in twenty that goes badly. **Registered:** the pass rule was written before the result was seen. **[a, b]:** the 95% interval.

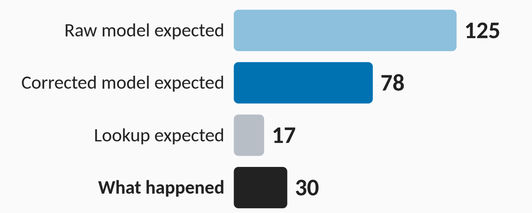
How it works — **1 Predict** which plants get built, what each asset produces → **2 Simulate** 2,000 futures of prices, output, cost and timing → **3 Optimize** the book, under the budget and the risk appetite → **4 Challenge** a governed language model argues with the committee → **5 Decide** a named person signs → **6 Score** on 2,000 different futures

1 STAGE 1 · PREDICTION

Can a model predict which plants get built?

It ranks well, but sizes badly.

Requests that started operating in 2025, of 1,441 live



One year after the decision; nothing refitted (report Figure 3).

- The model ordered requests better than the lookup: AUC 0.731 against 0.686.
- It put ERCOT's five-year chance at 0.380; 0.238 operated.
- Machine learning won 3 of 12 forecast contests, all for plant output; prices behaved as random walks.

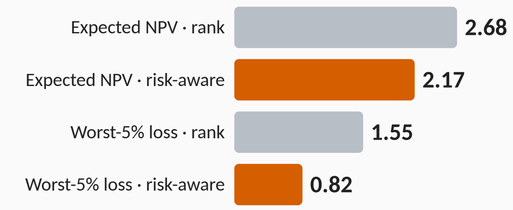
+$37M what the model's curves added to the book: [−$76M, +$151M], **not clear of zero**

2 STAGES 2–3 · RISK

Is the highest expected return the best investment?

No. Pricing the tail beat ranking by value.

Rank by value vs the risk-aware optimizer, $ billion



2,000 held-out futures (report Figure 4, Table 1).

- The tail is in the co-movement: five leases on one oil price, eight plants on one power price.
- The optimizer gave up $508M of expected value to nearly halve the worst-5% loss.
- It found a hedge ranking never would: five cheap wind and solar deposits that pay when the leases do not.

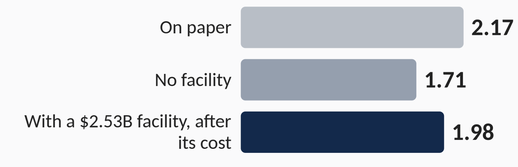
+$219M over ranking by value [+$13M, +$401M]. Registered (C33): it holds, narrowly

3 STAGE 3 · FINANCING

Can the company pay for what it chose?

Only with a credit line: choose the book and its financing together.

The risk-aware book's expected NPV, $ billion



Fee 0.5% and interest 8% are assumed terms (report Section 5, Appendix K.4).

- The construction reserve is an average: in half the futures the plants call more than it.
- The risk-aware book is the best of five ways to choose in 7 of 7 modelled worlds.
- **Caveat:** with no facility at all, equal capital by sector wins ($0.87B against $0.76B).

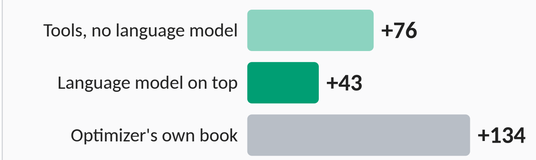
+$312M over ranking once every book is financed the same way [+$110M, +$506M]

4 STAGE 4 · AI CHALLENGER

Does a language model help the committee decide?

Yes, as a challenger that argues, not as the one who decides.

What each step adds to the committee's objective, $M per case



48 simulated committee proposals; from $863M to $1,116M per case (report Table 2).

- It named 12 of 12 flaws of argument, challenged 0 of 12 sound proposals and read 40 of 40 paraphrased rules exactly.
- A fixed optimizer-first rule chose books as well (−$14M): its value is the argument, not the pick.
- It failed 2 registered lines (8 of 48 readings missed the tail weight; 15 of 20 injections held); the controls caught both. A named person signs.

+$43M per case from the language model [+$9M, +$86M]. Registered (C34)

What the committee should do on 1 January 2025

1. State the tail weight first; an unstated one is read as zero.
2. Fund the risk-aware book: 13 of 40 candidates, $3.78B. Invest now in 8, in stages in 5.
3. Arrange a $2.53B contingent construction facility.
4. Wait on 12 requests, monitor 14 leases, reject 1. Run the challenger before signing.

Beyond prediction lies a discipline

- Simulate what you cannot forecast.
- Optimize against the risk you are willing to bear.
- Price the capital you will need.
- Let AI argue with the decision-maker, not replace them.

Read with care

Modelled, not realized returns. The committees are simulated, the lease costs assumed, the futures resample 2018–2024, and the primary result clears zero by only $13M. 41 criteria and 87 lines were registered; 40 pass, and every failure is kept as scored.

Data: LBNL **Queued Up** 2025 and 2026 editions (Rand et al., 2025, 2026), ERCOT interconnection reports and hourly prices, BSEE lease production, EIA prices and generation, NREL ATB 2024; author's analysis. Methods: LightGBM discrete-time hazard models; block-bootstrap futures (Künsch, 1989); a mixed-integer program on expected NPV minus the worst-5% loss (Rockafellar & Uryasev, 2000). Numbers from the report, Version 2; each traces to `results/report_numbers.csv`. Code: github.com/alanmossinger/Capstone